Reproducing GitHub Insights for Diffusers

26AI60R04
CD61203: Essential Tools for Scientific Computing

13 November 2026

The four figures reproduce charts from the public repository huggingface/diffusers. References were captured from GitHub on 13 November 2026, in the light theme. The data, plotted values, source URLs, capture times, and browser settings are retained in the repository. Each reproduction is drawn with Python/Matplotlib from CSV, independently of the reference screenshot.

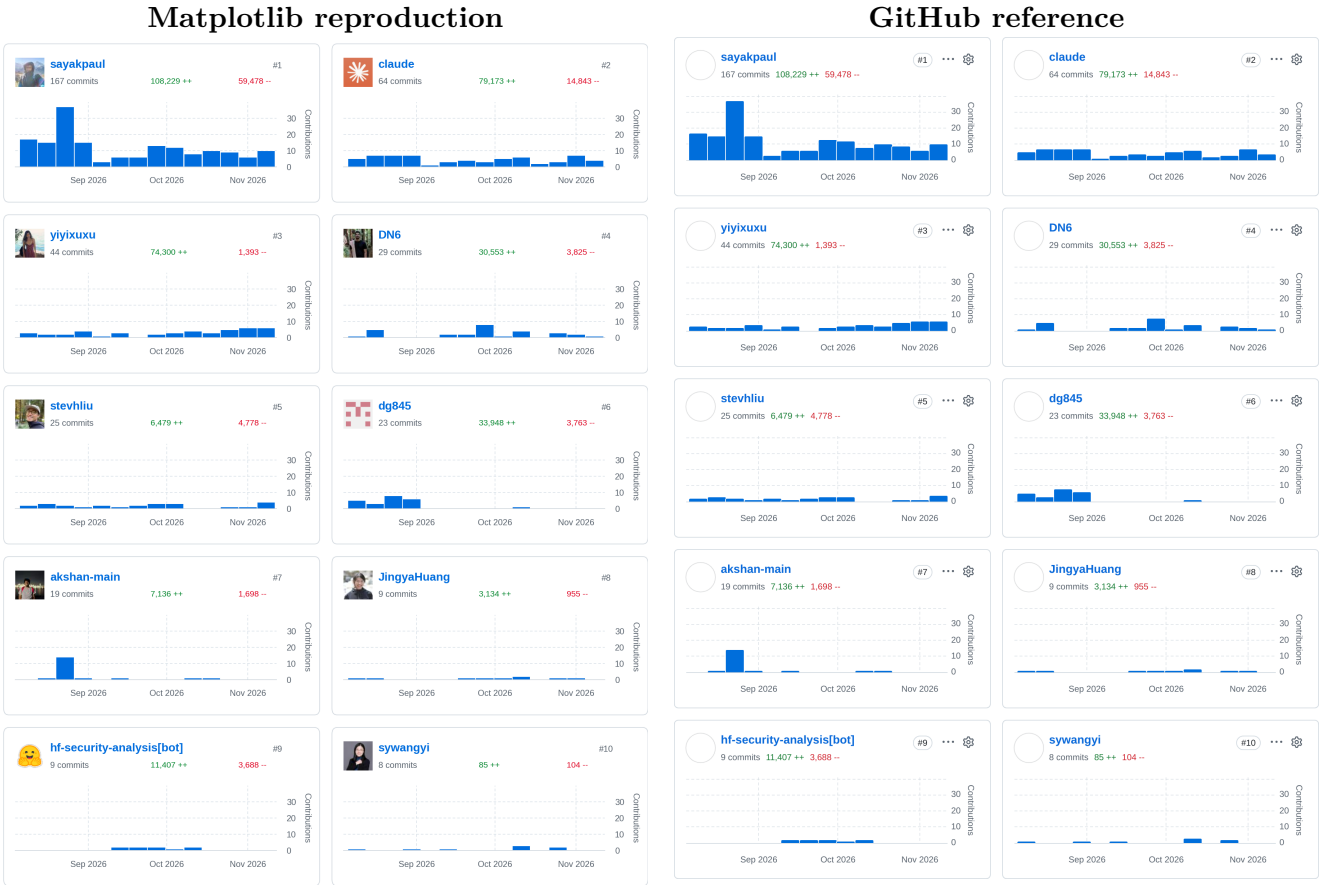
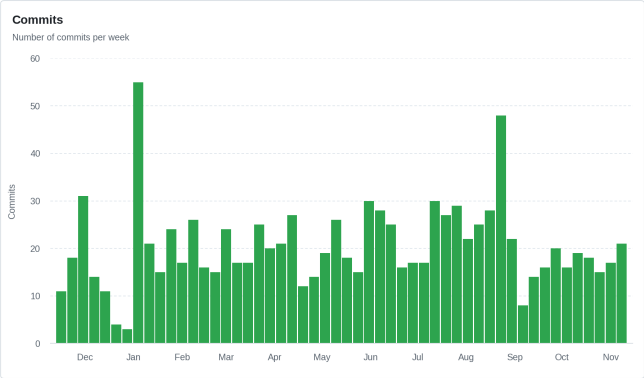


Figure 1: Contributors: the first ten ranked contributor cards for the default "Last 3 months" selection, with 14 weekly bars per author, from 9 August to 8 November 2026. Blue bars show commits to `main`, excluding merge commits. The right-hand axes use the common scale of the reference. Names, rank, commit totals, additions, and deletions follow GitHub's displayed order, including ties. The source contains additional contributors; the matching reference region and reproduction both show the first ten.

Source: https://github.com/huggingface/diffusers/graphs/contributors.

Weekly commits

Matplotlib reproduction



GitHub reference

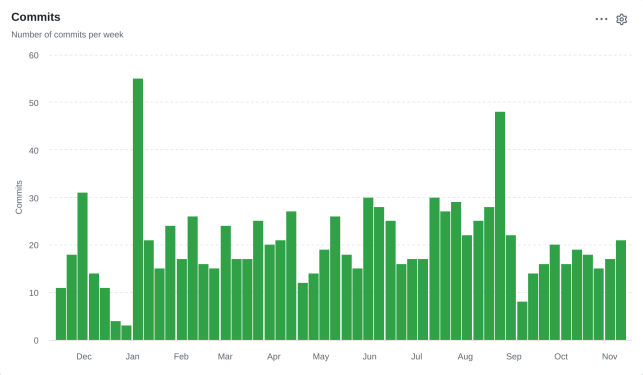


Figure 2: Commits over the last year: 52 weekly totals from 16 November 2025 through 8 November 2026. The green column chart preserves the month labels, the "Commits" axis label, and the 0–60 vertical scale. The largest weekly total is 55, in the week beginning 4 January 2026.

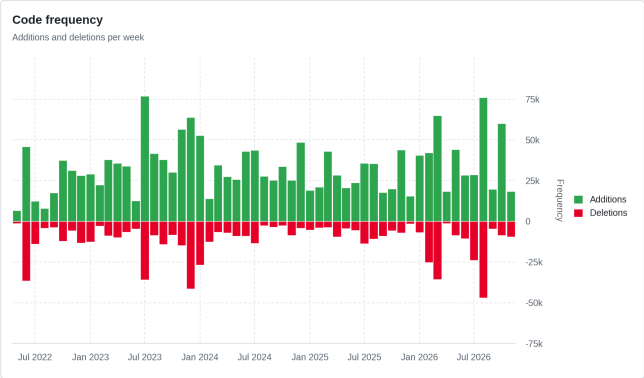
Source: https://github.com/huggingface/diffusers/graphs/commit-activity.

Data and method. The GitHub chart endpoint supplies a total and seven daily counts for every week. The CSV stores all eight counts and the UTC week-start date. The figure uses the supplied weekly total without smoothing or interpolation. As an independent consistency check, every total equals the sum of its daily counts.

Comparison. All 52 values agree exactly with the accessible point labels captured from GitHub's rendered chart. Font rendering, rounded corners, and interactive controls are not required to be pixel-identical.

Code frequency

Matplotlib reproduction



GitHub reference

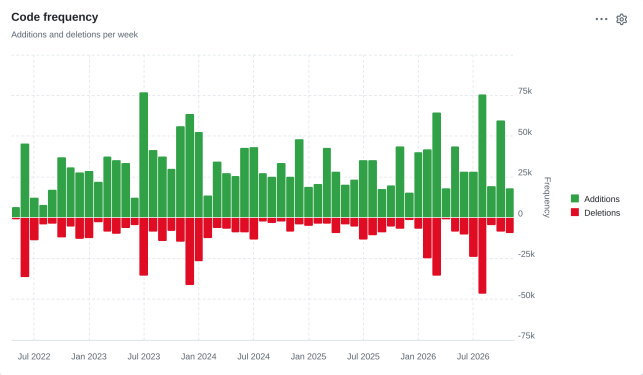


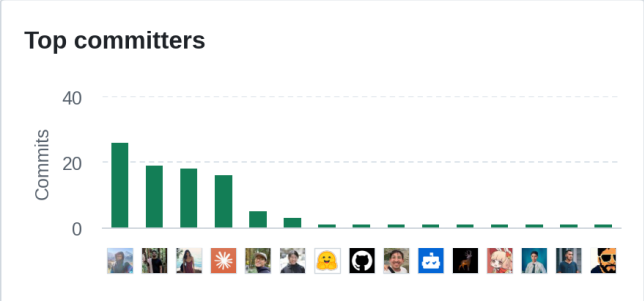
Figure 3: Code frequency across repository history. Green columns are additions; red columns are signed deletions below zero. The raw endpoint provides 233 weekly observations, from 29 May 2022 to 8 November 2026. At the captured full-history view GitHub groups those observations into 55 calendar-month sums; the reproduction uses those same sums to match the visible reference. The title and subtitle retain GitHub's wording, including "per week".

Source: https://github.com/huggingface/diffusers/graphs/code-frequency.

Aggregation. Each weekly observation is assigned to the calendar month containing its UTC week-start date. Additions and signed deletions are summed separately. The weekly source remains in `raw_data.csv`; the monthly values are also exported as `displayed_data.csv`. Plotting code recomputes the sums from the weekly source. All 110 displayed series values agree exactly with the captured GitHub point labels. The largest monthly additions count is 76,969; the most negative monthly deletions count is $-46,931$.

Top committers

Matplotlib reproduction



GitHub reference

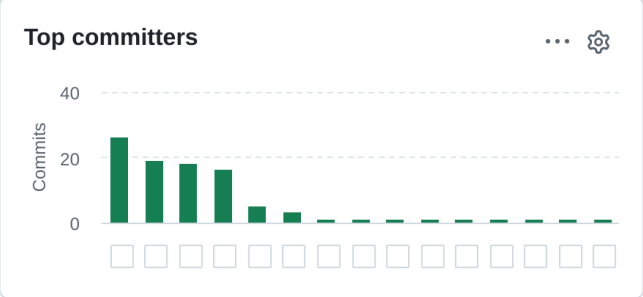


Figure 4: The actual "Top committers" chart in GitHub Insights/Pulse for the displayed period 5–12 November 2026. All 15 authors appear in the original order. The green columns, 0–40 "Commits" scale, and avatar category labels match the reference. The leading counts are 26 for sayakpaul, 19 for DN6, 18 for yiyixuxu, and 16 for claude.

Source: `https://github.com/huggingface/diffusers/pulse`.

Scope. This chart is located inside Pulse rather than in a separate Top committers menu entry. Its period differs from the Contributors page. Values are taken directly from the rendered chart's accessible point labels; author names are retained in the CSV even though GitHub displays avatars on the category axis. Pulse's prose summary reports separate counts excluding merges; those summary counts are not substituted for the plotted author counts.

Reproducibility and validation. Run `python reproduce.py` to regenerate all figures offline, and `python tools/validate.py` to compare all 317 plotted values against the saved GitHub reference labels. Run `pdflatex main.tex` twice, or `tectonic main.tex`, to build this document. The comparisons in Figures 1–4 use authentic browser screenshots of the corresponding chart regions. No synthetic reference images or invented data are used.